ANSWER KEY & MARKING SCHEME SET 1

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 23	$\frac{4325+25p}{77+p}=49$ gives $(25-49)p=-552$; $p=23$
2	**(c)** 9	$238.4=230+\frac{50-(20+x)}{25}\times 10$ gives $x=9$; $x=9$; then $y=100-82-x=9$
3	**(b)** 269.8	$\sum f_i=100$ and $\sum f_ix_i=26980$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{26980}{100}=269.8$
4	**(c)** 108	Median class 100–120: $l=100$, $cf=44$, $f=15$, $h=20$; Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=100+\frac{50-44}{15}\times 20=108$
5	**(a)** 63	$l=60$, $f_1=24$, $f_0=21$, $f_2=17$, $h=10$; Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=60+\frac{3}{10}\times 10=63$
6	**(a)** 25	$20+\frac{25-18}{14}\times 10=25$
7	**(a)** 20	$\frac{14660+190p}{80+p}=184.6$ gives $(190-184.6)p=108$; $p=20$
8	**(b)** 21	$25=20+\frac{50-(20+x)}{18}\times 10$ gives $x=21$; $x=21$; then $y=100-75-x=4$
9	**(b)** 71	$\sum f_i=100$ and $\sum f_ix_i=7100$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{7100}{100}=71$
10	**(c)** 190	Median class 180–200: $l=180$, $cf=39$, $f=22$, $h=20$; Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=180+\frac{50-39}{22}\times 20=190$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\sum f_i=83+p$ and $\sum f_ix_i=23150+250p$; $\frac{23150+250p}{83+p}=274$ gives $(250-274)p=-408$ (1); $p=17$ (1).	1+1
12	Highest frequency is 23, so the modal class is 20–30; $l=20$, $f_1=23$, $f_0=4$, $f_2=4$, $h=10$ (1); Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=20+\frac{19}{38}\times 10=25$ (1).	1+1
13	Cumulative frequencies: 23, 27, 52, 58, 61, 80, 100; $N=100$, so $\frac{N}{2}=50$; Median class 240–260: $l=240$, $cf=27$, $f=25$, $h=20$ (1); Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=240+\frac{50-27}{25}\times 20=258.4$ (1).	1+1
14	$\sum f_i=98+p$ and $\sum f_ix_i=7730+75p$; $\frac{7730+75p}{98+p}=78.8$ gives $(75-78.8)p=-7.6$ (1); $p=2$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sum f_i=100$ and $\sum f_ix_i=7530$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{7530}{100}=75.3$ (1); $l=80$, $f_1=23$, $f_0=20$, $f_2=2$, $h=10$ (1); Mode: $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=80+\frac{3}{24}\times 10=81.25$ (1).	1+1+1
16	Take $a=75$, $h=10$; $u_i=\frac{x_i-a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i=100$ and $\sum f_iu_i=-70$ (1); Mean $=a+h\frac{\sum f_iu_i}{\sum f_i}=75+10\times\frac{-70}{100}=68$ (1).	1+1+1
17	Median class is 230–240 with $f=20$, and the cumulative frequency before it is $cf=24+x$ (1); $239.5=230+\frac{50-(24+x)}{20}\times 10$ gives $x=7$ (1); $x=7$; then $y=100-80-x=13$ (1).	1+1+1
	OR: $\sum f_i=100$ and $\sum f_ix_i=10360$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{10360}{100}=103.6$ (1); $l=120$, $f_1=24$, $f_0=15$, $f_2=17$, $h=20$ (1); Mode: $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=120+\frac{9}{16}\times 20=131.25$ (1).	1+1+1
18	Take $a=270$, $h=20$; $u_i=\frac{x_i-a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i=100$ and $\sum f_iu_i=-28$ (1); Mean $=a+h\frac{\sum f_iu_i}{\sum f_i}=270+20\times\frac{-28}{100}=264.4$ (1).	1+1+1
	OR: Median class is 50–60 with $f=24$, and the cumulative frequency before it is $cf=17+x$ (1); $55=50+\frac{50-(17+x)}{24}\times 10$ gives $x=21$ (1); $x=21$; then $y=100-56-x=23$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Median class is 230–240 with $f=25$, and the cumulative frequency before it is $cf=20+x$ (2); $238.4=230+\frac{50-(20+x)}{25}\times 10$ gives $x=9$ (2); $x=9$; then $y=100-82-x=9$ (1).	2+2+1
	OR: $\sum f_i=100$ and $\sum f_ix_i=7880$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{7880}{100}=78.8$ (2); $l=50$, $f_1=23$, $f_0=7$, $f_2=19$, $h=10$ (2); Mode: $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=50+\frac{16}{20}\times 10=58$ (1).	2+2+1

20	Median class is 20–30 with $f = 18$, and the cumulative frequency before it is $cf = 20 + x$ (2); $25 = 20 + \frac{50-(20+x)}{18} \times 10$ gives $x = 21$ (2); $x = 21$; then $y = 100 - 75 - x = 4$ (1).	2+2+1
	OR: $\sum f_i = 100$ and $\sum f_i x_i = 5530$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{5530}{100} = 55.3$ (2); $l = 60$, $f_1 = 23$, $f_0 = 16$, $f_2 = 10$, $h = 10$ (2); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{7}{20} \times 10 = 63.5$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 242	$l=240$, $f_1=24$, $f_0=22$, $f_2=6$, $h=20$; Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=240+\frac{2}{20}\times 20=242$
2	**(b)** $20+\frac{10}{18}\times 10$	$\frac{f_1-f_0}{2f_1-f_0-f_2}=\frac{10}{32-14}$
3	**(d)** 25	$\frac{7890+110p}{75+p}=106.4$ gives $(110-106.4)p=90$; $p=25$
4	**(a)** 22	$70=60+\frac{50-(20+x)}{16}\times 20$ gives $x=22$; $x=22$; then $y=100-69-x=9$
5	**(a)** 74.4	$\sum f_i=100$ and $\sum f_ix_i=7440$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{7440}{100}=74.4$
6	**(a)** 65	Median class 60–70: $l=60$, $cf=49$, $f=2$, $h=10$; Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=60+\frac{50-49}{2}\times 10=65$
7	**(c)** 234	$l=230$, $f_1=24$, $f_0=18$, $f_2=15$, $h=10$; Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=230+\frac{6}{15}\times 10=234$
8	**(d)** 43	$45+10\times\frac{-12}{60}$
9	**(a)** 15	$\frac{15990+210p}{85+p}=191.4$ gives $(210-191.4)p=279$; $p=15$
10	**(d)** 10	$56=50+\frac{50-(28+x)}{20}\times 10$ gives $x=10$; $x=10$; then $y=100-82-x=8$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Highest frequency is 22, so the modal class is 220–240; $l=220$, $f_1=22$, $f_0=13$, $f_2=7$, $h=20$ (1); Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=220+\frac{9}{24}\times 20=227.5$ (1).	1+1
12	Cumulative frequencies: 21, 34, 59, 64, 70, 94, 98, 100; $N=100$, so $\frac{N}{2}=50$; Median class 60–70: $l=60$, $cf=34$, $f=25$, $h=10$ (1); Median $=l+\frac{\frac{N}{2}-cf}{f}\times h=60+\frac{50-34}{25}\times 10=66.4$ (1).	1+1
13	$\sum f_i=77+p$ and $\sum f_ix_i=4035+65p$; $\frac{4035+65p}{77+p}=55.3$ gives $(65-55.3)p=223.1$ (1); $p=23$ (1).	1+1
14	Highest frequency is 25, so the modal class is 100–120; $l=100$, $f_1=25$, $f_0=13$, $f_2=5$, $h=20$ (1); Mode $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=100+\frac{12}{32}\times 20=107.5$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sum f_i=100$ and $\sum f_ix_i=18680$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{18680}{100}=186.8$ (1); $l=180$, $f_1=22$, $f_0=21$, $f_2=21$, $h=20$ (1); Mode: $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=180+\frac{1}{2}\times 20=190$ (1).	1+1+1
16	Take $a=245$, $h=10$; $u_i=\frac{x_i-a}{h}$: -4, -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i=100$ and $\sum f_iu_i=-33$ (1); Mean $=a+h\frac{\sum f_iu_i}{\sum f_i}=245+10\times\frac{-33}{100}=241.7$ (1).	1+1+1
17	Median class is 80–100 with $f=16$, and the cumulative frequency before it is $cf=31+x$ (1); $93.75=80+\frac{50-(31+x)}{16}\times 20$ gives $x=8$ (1); $x=8$; then $y=100-76-x=16$ (1).	1+1+1
	OR: $\sum f_i=100$ and $\sum f_ix_i=6380$; Mean $=\frac{\sum f_ix_i}{\sum f_i}=\frac{6380}{100}=63.8$ (1); $l=80$, $f_1=23$, $f_0=16$, $f_2=10$, $h=10$ (1); Mode: $=l+\frac{f_1-f_0}{2f_1-f_0-f_2}\times h=80+\frac{7}{20}\times 10=83.5$ (1).	1+1+1
18	Take $a=270$, $h=20$; $u_i=\frac{x_i-a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i=100$ and $\sum f_iu_i=50$ (1); Mean $=a+h\frac{\sum f_iu_i}{\sum f_i}=270+20\times\frac{50}{100}=280$ (1).	1+1+1
	OR: Median class is 250–260 with $f=24$, and the cumulative frequency before it is $cf=41+x$ (1); $252.5=250+\frac{50-(41+x)}{24}\times 10$ gives $x=3$ (1); $x=3$; then $y=100-80-x=17$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Median class is 60–80 with $f=16$, and the cumulative frequency before it is $cf=20+x$ (2); $70=60+\frac{50-(20+x)}{16}\times 20$ gives $x=22$ (2); $x=22$; then $y=100-69-x=9$ (1).	2+2+1

	OR: $\sum f_i = 100$ and $\sum f_i x_i = 10760$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{10760}{100} = 107.6$ (2); $l = 60$, $f_1 = 25$, $f_0 = 16$, $f_2 = 16$, $h = 20$ (2); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 60 + \frac{9}{18} \times 20 = 70$ (1).	2+2+1
20	$\sum f_i = 89 + p$ and $\sum f_i x_i = 9590 + 70p$ (2); $\frac{9590+70p}{89+p} = 103.6$ gives $(70 - 103.6)p = -369.6$ (2); $p = 11$ (1).	2+2+1
	OR: Take assumed mean $a = 45$; deviations $d_i = x_i - a$: -40, -30, -20, -10, 0, 10, 20, 30 (2); $\sum f_i = 100$ and $\sum f_i d_i = 400$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 45 + \frac{400}{100} = 49$ (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 13: **Statistics** | Class 10 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 77	$\sum f_i = 100$ and $\sum f_i x_i = 7700$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7700}{100} = 77$
2	**(b)** 96.25	Median class 80–100: $l = 80$, $cf = 37$, $f = 16$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 80 + \frac{50-37}{16} \times 20 = 96.25$
3	**(d)** 231.25	$l = 220$, $f_1 = 21$, $f_0 = 3$, $f_2 = 7$, $h = 20$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 220 + \frac{18}{32} \times 20 = 231.25$
4	**(d)** 4	$\bar{x} - a$
5	**(d)** 25	$\frac{8130+70p}{75+p} = 98.8$ gives $(70 - 98.8)p = -720$; $p = 25$
6	**(d)** 14	$290 = 280 + \frac{50-(28+x)}{16} \times 20$ gives $x = 14$; $x = 14$; then $y = 100 - 66 - x = 20$
7	**(c)** 75.3	$\sum f_i = 100$ and $\sum f_i x_i = 7530$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7530}{100} = 75.3$
8	**(b)** 68.75	Median class 60–80: $l = 60$, $cf = 43$, $f = 16$, $h = 20$; Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 60 + \frac{50-43}{16} \times 20 = 68.75$
9	**(a)** 23	$l = 20$, $f_1 = 25$, $f_0 = 22$, $f_2 = 8$, $h = 20$; Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 20 + \frac{3}{20} \times 20 = 23$
10	**(a)** 45	$40 + \frac{30-22}{16} \times 10$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Cumulative frequencies: 19, 30, 38, 63, 71, 85, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 70–80: $l = 70$, $cf = 38$, $f = 25$, $h = 10$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 70 + \frac{50-38}{25} \times 10 = 74.8$ (1).	1+1
12	$\sum f_i = 95 + p$ and $\sum f_i x_i = 10210 + 110p$; $\frac{10210+110p}{95+p} = 107.6$ gives $(110 - 107.6)p = 12$ (1); $p = 5$ (1).	1+1
13	Highest frequency is 20, so the modal class is 180–200; $l = 180$, $f_1 = 20$, $f_0 = 9$, $f_2 = 15$, $h = 20$ (1); Mode $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 180 + \frac{11}{16} \times 20 = 193.75$ (1).	1+1
14	Cumulative frequencies: 10, 17, 25, 48, 73, 92, 100; $N = 100$, so $\frac{N}{2} = 50$; Median class 80–100: $l = 80$, $cf = 48$, $f = 25$, $h = 20$ (1); Median $= l + \frac{\frac{N}{2} - cf}{f} \times h = 80 + \frac{50-48}{25} \times 20 = 81.6$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sum f_i = 100$ and $\sum f_i x_i = 8840$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{8840}{100} = 88.4$ (1); $l = 100$, $f_1 = 25$, $f_0 = 16$, $f_2 = 9$, $h = 20$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 100 + \frac{9}{25} \times 20 = 107.2$ (1).	1+1+1
16	Take $a = 70$, $h = 20$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i = 100$ and $\sum f_i u_i = -71$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 70 + 20 \times \frac{-71}{100} = 55.8$ (1).	1+1+1
17	Median class is 120–140 with $f = 16$, and the cumulative frequency before it is $cf = 23 + x$ (1); $127.5 = 120 + \frac{50-(23+x)}{16} \times 20$ gives $x = 21$ (1); $x = 21$; then $y = 100 - 62 - x = 17$ (1).	1+1+1
	OR: $\sum f_i = 100$ and $\sum f_i x_i = 7620$; Mean $= \frac{\sum f_i x_i}{\sum f_i} = \frac{7620}{100} = 76.2$ (1); $l = 120$, $f_1 = 24$, $f_0 = 5$, $f_2 = 5$, $h = 20$ (1); Mode: $= l + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h = 120 + \frac{19}{38} \times 20 = 130$ (1).	1+1+1
18	Take $a = 270$, $h = 20$; $u_i = \frac{x_i - a}{h}$: -3, -2, -1, 0, 1, 2, 3 (1); $\sum f_i = 100$ and $\sum f_i u_i = 20$ (1); Mean $= a + h\frac{\sum f_i u_i}{\sum f_i} = 270 + 20 \times \frac{20}{100} = 274$ (1).	1+1+1
	OR: Median class is 260–280 with $f = 12$, and the cumulative frequency before it is $cf = 37 + x$ (1); $265 = 260 + \frac{50-(37+x)}{12} \times 20$ gives $x = 10$ (1); $x = 10$; then $y = 100 - 72 - x = 18$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$\sum f_i = 79 + p$ and $\sum f_i x_i = 15110 + 170p$ (2); $\frac{15110+170p}{79+p} = 186.8$ gives $(170 - 186.8)p = -352.8$ (2); $p = 21$ (1).	2+2+1

	OR: Take assumed mean $a = 190$; deviations $d_i = x_i - a$: -80, -60, -40, -20, 0, 20, 40, 60 (2); $\sum f_i = 100$ and $\sum f_i d_i = -540$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 190 + \frac{-540}{100} = 184.6$ (1).	2+2+1
20	$\sum f_i = 98 + p$ and $\sum f_i x_i = 6250 + 65p$ (2); $\frac{6250+65p}{98+p} = 63.8$ gives $(65 - 63.8)p = 2.4$ (2); $p = 2$ (1).	2+2+1
	OR: Take assumed mean $a = 130$; deviations $d_i = x_i - a$: -80, -60, -40, -20, 0, 20, 40, 60 (2); $\sum f_i = 100$ and $\sum f_i d_i = -2360$ (2); Mean $= a + \frac{\sum f_i d_i}{\sum f_i} = 130 + \frac{-2360}{100} = 106.4$ (1).	2+2+1

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